

Practical**SECTION 1 : DDL (DATA DEFINITION LANGUAGE)****QUESTION:1 - CREATE TABLE**

```
mysql> CREATE DATABASE bankdb;
Query OK, 1 row affected (0.08 sec)

mysql> USE bankdb;
Database changed
mysql> CREATE TABLE bank_transactions (
    ->     txn_id INT PRIMARY KEY,
    ->     customer_name VARCHAR(50),
    ->     branch_name VARCHAR(50),
    ->     transaction_type VARCHAR(20),
    ->     amount DECIMAL(10,2),
    ->     transaction_date DATE
    -> );
Query OK, 0 rows affected (0.07 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_bankdb |
+-----+
| bank_transactions |
+-----+
1 row in set (0.02 sec)
```

QUESTION 2 - ALTER TABLE ADD COLUMN

```
mysql> ALTER TABLE bank_transactions
    -> ADD account_no VARCHAR(20);
Query OK, 0 rows affected (0.09 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> DESCRIBE bank_transactions;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| txn_id         | int           | NO   | PRI | NULL    |       |
| customer_name  | varchar(50)   | YES  |     | NULL    |       |
| branch_name    | varchar(50)   | YES  |     | NULL    |       |
| transaction_type | varchar(20)   | YES  |     | NULL    |       |
| amount         | decimal(10,2) | YES  |     | NULL    |       |
| transaction_date | date          | YES  |     | NULL    |       |
| account_no     | varchar(20)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.02 sec)
```

QUESTION 3 - ALTER TABLE MODIFY COLUMN

```
mysql> ALTER TABLE bank_transactions
-> MODIFY customer_name VARCHAR(100);
Query OK, 0 rows affected (0.06 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> DESCRIBE bank_transactions;
```

Field	Type	Null	Key	Default	Extra
txn_id	int	NO	PRI	NULL	
customer_name	varchar(100)	YES		NULL	
branch_name	varchar(50)	YES		NULL	
transaction_type	varchar(20)	YES		NULL	
amount	decimal(10,2)	YES		NULL	
transaction_date	date	YES		NULL	
account_no	varchar(20)	YES		NULL	

```
7 rows in set (0.00 sec)
```

QUESTION 4 - RENAME TABLE

```
mysql> RENAME TABLE bank_transactions
-> TO customer_transactions;
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> SHOW TABLES;
```

Tables_in_bankdb
customer_transactions

```
1 row in set (0.00 sec)
```

QUESTION 5 - TRUNCATE TABLE

```
mysql> CREATE TABLE customer_transactions (  
->     txn_id INT PRIMARY KEY,  
->     customer_name VARCHAR(100),  
->     branch_name VARCHAR(50),  
->     transaction_type VARCHAR(20),  
->     amount DECIMAL(10,2),  
->     transaction_date DATE,  
->     account_no VARCHAR(20)  
-> );
```

Query OK, 0 rows affected (0.02 sec)

```
mysql> TRUNCATE TABLE customer_transactions;  
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> SELECT * FROM customer_transactions;  
Empty set (0.00 sec)
```

```
mysql> SHOW TABLES;
```

```
+-----+  
| Tables_in_bankdb |  
+-----+  
| customer_transactions |  
+-----+  
1 row in set (0.00 sec)
```

QUESTION 6 - DROP TABLE

```
mysql> DROP TABLE customer_transactions;  
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SHOW TABLES;  
Empty set (0.00 sec)
```

SECTION 2 : DML (DATA MANIPULATION LANGUAGE)

QUESTION 7 - INSERT

```
mysql> INSERT INTO customer_transactions
  -> VALUES
  -> (111,'Venu','Vizag','Deposit',7000,'2024-01-18','ACC111');
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM customer_transactions;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 111 | Venu | Vizag | Deposit | 7000.00 | 2024-01-18 | ACC111 |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

QUESTION 8 - UPDATE

```
mysql> UPDATE customer_transactions
  -> SET amount = 5000
  -> WHERE txn_id = 105;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 1 Changed: 0 Warnings: 0

mysql> SELECT * FROM customer_transactions
  -> WHERE txn_id = 105;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 105 | Priya | Hyderabad | Withdrawal | 5000.00 | 2024-01-11 | ACC105 |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

QUESTION 9 - DELETE

```
mysql> DELETE FROM customer_transactions
  -> WHERE txn_id = 111;
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM customer_transactions
  -> WHERE txn_id = 111;
Empty set (0.01 sec)
```

QUESTION 10 - SELECT

```
mysql> SELECT * FROM customer_transactions;
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05	ACC101
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06	ACC102
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	ACC103
104	Anil	Vizag	Deposit	8000.00	2024-01-10	ACC104
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11	ACC105
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12	ACC106
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	ACC107
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	ACC108
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	ACC109
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	ACC110

10 rows in set (0.00 sec)

QUESTION 11 - SELECT WITH WHERE

```
mysql> SELECT *
-> FROM customer_transactions
-> WHERE transaction_type = 'Deposit';
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05	ACC101
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	ACC103
104	Anil	Vizag	Deposit	8000.00	2024-01-10	ACC104
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12	ACC106
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	ACC108
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	ACC110

6 rows in set (0.00 sec)

Question 12 – SELECT with ORDER BY

```
mysql> SELECT *
-> FROM customer_transactions
-> ORDER BY amount DESC;
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12	ACC106
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	ACC103
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	ACC110
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	ACC108
104	Anil	Vizag	Deposit	8000.00	2024-01-10	ACC104
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05	ACC101
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11	ACC105
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	ACC109
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06	ACC102
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	ACC107

10 rows in set (0.00 sec)

SECTION 3 : DCL (DATA CONTROL LANGUAGE)**Question 13 – GRANT SELECT**

```
mysql> CREATE USER 'Auditor1'@'localhost'
      -> IDENTIFIED BY 'password123';
Query OK, 0 rows affected (0.03 sec)

mysql> GRANT SELECT
      -> ON customer_transactions
      -> TO 'Auditor1'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.01 sec)
```

Question 14 – GRANT ALL PRIVILEGES

```
mysql> CREATE USER 'BranchManager'@'localhost'
      -> IDENTIFIED BY 'password123';
Query OK, 0 rows affected (0.01 sec)

mysql> GRANT ALL PRIVILEGES
      -> ON customer_transactions
      -> TO 'BranchManager'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.01 sec)
```

Question 15 – REVOKE SELECT

```
mysql> REVOKE SELECT
      -> ON customer_transactions
      -> FROM 'Auditor1'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> SHOW GRANTS FOR 'Auditor1'@'localhost';
+-----+
| Grants for Auditor1@localhost |
+-----+
| GRANT USAGE ON *.* TO `Auditor1`@`localhost` |
+-----+
1 row in set (0.00 sec)
```

Question 16 – REVOKE ALL PRIVILEGES

```
mysql> REVOKE ALL PRIVILEGES
      -> ON customer_transactions
      -> FROM 'BranchManager'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> SHOW GRANTS FOR 'BranchManager'@'localhost';
+-----+
| Grants for BranchManager@localhost |
+-----+
| GRANT USAGE ON *.* TO `BranchManager`@`localhost` |
+-----+
1 row in set (0.00 sec)
```

SECTION 4 : TCL (TRANSACTION CONTROL LANGUAGE)

Question 17 – COMMIT

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO customer_transactions
      -> VALUES
      -> (112,'Asha','Hyderabad','Deposit',6000,'2024-01-20','ACC112');
Query OK, 1 row affected (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM customer_transactions
      -> WHERE txn_id = 112;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 112 | Asha | Hyderabad | Deposit | 6000.00 | 2024-01-20 | ACC112 |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

Question 18 – ROLLBACK

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO customer_transactions
      -> VALUES
      -> (113,'Rohan','Vizag','Deposit',8000,'2024-01-21','ACC113');
Query OK, 1 row affected (0.00 sec)

mysql> ROLLBACK;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM customer_transactions
      -> WHERE txn_id = 113;
Empty set (0.00 sec)
```

QUESTION 19 - ROLLBACK TO SAVEPOINT

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO customer_transactions
-> VALUES
-> (114,'Kiran','Hyderabad','Deposit',9000,'2024-01-22','ACC114');
Query OK, 1 row affected (0.00 sec)

mysql> SAVEPOINT sp1;
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO customer_transactions
-> VALUES
-> (115,'Meena','Vizag','Withdrawal',2500,'2024-01-23','ACC115');
Query OK, 1 row affected (0.00 sec)

mysql> ROLLBACK TO sp1;
Query OK, 0 rows affected (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM customer_transactions;
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05	ACC101
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06	ACC102
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	ACC103
104	Anil	Vizag	Deposit	8000.00	2024-01-10	ACC104
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11	ACC105
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12	ACC106
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	ACC107
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	ACC108
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	ACC109
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	ACC110
112	Asha	Hyderabad	Deposit	6000.00	2024-01-20	ACC112
114	Kiran	Hyderabad	Deposit	9000.00	2024-01-22	ACC114

```
12 rows in set (0.00 sec)
```

QUESTION 20 - FULL TCL EXAMPLE

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO customer_transactions
-> VALUES
-> (116,'Anita','Vijayawada','Deposit',10000,'2024-01-24','ACC116');
Query OK, 1 row affected (0.00 sec)

mysql> SAVEPOINT sp2;
Query OK, 0 rows affected (0.00 sec)

mysql> RELEASE SAVEPOINT sp2;
Query OK, 0 rows affected (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM customer_transactions
-> WHERE txn_id = 116;
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
116	Anita	Vijayawada	Deposit	10000.00	2024-01-24	ACC116

```
1 row in set (0.00 sec)
```


